Acid leaching experiments of silicate rocks and its implications for mobility of cations during silicate weathering and dissolution of extra-terrestrial samples

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Declaration

I, **Rishant Prakash**, with SR No. 05-06-01-10-51-21-1-19647, hereby affirm that the thesis titled, **'Acid leaching experiments of silicate rocks and its implications for mobility of cations during silicate weathering and dissolution of extra-terrestrial samples'** represents an original work conducted under the supervision of Prof Ramananda Chakrabarti within the RC-Lab Group at the Centre for Earth Sciences, Indian Institute of Science, Bangalore. The research was undertaken during my post-graduate tenure, pursuing a Master of Technology in Earth Science at the Indian Institute of Science from August 2021 to May 2023. In instances where I have incorporated ideas or findings from external sources, appropriate acknowledgement has been duly provided by including references.

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In my capacity as supervisor of the above-mentioned work, I certify that the above statements are true to the best of my knowledge.

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Abstract

Chemical weathering of silicate rocks and minerals result in net removal of atmospheric CO2 over geological timescales and controls Earth's surface temperatures over time. Kinetics of silicate weathering is sensitive to temperature, CO2 emission, and land surface organisation. However, there is much debate over its response strength and evolution. Among all silicate rocks, basalt display a high weathering rate and contributes to a large part of the CO2 consumption flux. Dissolution of silicate rocks and minerals in the laboratory require HF which is necessary to break the silica tetrahedra. However, simple equations of natural silicate weathering do not consider HF and typically consider carbonic acid or sulphuric acid. This suggests that natural silicate weathering is not congruent and hence, laboratory experiments of silicate dissolution without the use of HF are necessary. The objective of this thesis's first part is to understand the mobility of elements from silicate rocks comprising silicate minerals with different silicate structures (neso- to tectosilicate), which in turn controls the stability of the minerals, when reacted with non-HF acids. To expedite the reactions, the experiments were conducted at high pressure and temperature using a High-Pressure Asher.

The second part of this thesis is focussed on geochemical characterisation of the Sadiya meteorite, which is an Ordinary Chondrite. Ordinary Chondrites are undifferentiated and include the H, L, and LL groups which stand for High-Iron, Low-Iron, Low-Iron-Low-Metal, respectively. Sadiya is an observed recent fall which occurred on June 5, 2017, with a weight of 3 kg and density of 3.4 g/cm^3 (Saikia et al., 2017). The H, L, and LL groups are resolved using various parameters. Because of the presence of limited amount of sample in powdered form only bulk composition data has been measured in this study. As a result, "Sadiya" was characterised as H/L chondrite based on moderately siderophile elements. The volatile elements K and Na also show that Sadiya does not fit within ordinary chondrites' three groups (H, L, LL). Therefore, the possibility of a fractionating range of ordinary chondrites should be explored which will later accommodate the several meteorites that are currently grouped as H/L, L/LL etc. Moreover, determining petrographic type is important to understand a chondrite's thermal metamorphism or aqueous alteration history. However, it cannot be determined without petrographic and textural studies which will be conducted as part of future studies.

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Chapter 1
Introduction

1.1 Silicate weathering in nature:

Silicate weathering is a natural geological process that facilitates the removal of carbon dioxide (CO2) from the atmosphere and the subsequent storage of carbon in the Earth's crust. The process involves the reaction of CO2 with water to form carbonic acid, which then reacts with silicate minerals (ultramafic to felsic type) to produce dissolved ions, including bicarbonate, calcium, and magnesium (J. Gaillardet., 1999; Amundson et al., 2015). These ions are transported via water to the oceans, where marine organisms can use them to create shells and other structures.

Silicate weathering is influenced by various environmental factors, including rock type, soil type, climate, and vegetation cover (Raymond, P et al., 2013). It is carried out via acid dissolution, oxidation, hydrolysis, biological action, and spheroidal weathering over long periods (Berner, 2003). It is a crucial component of the global carbon cycle, as it helps regulate atmospheric CO_2 levels and the Earth's climate (Amundson et al., 2015).

Human activities, including deforestation and land use changes, have the potential to alter silicate weathering rates and the overall carbon cycle, with potential implications for climate change (Raymond, P et al., 2013; Amundson et al., 2015). Therefore, understanding the mechanisms and drivers of silicate weathering is important for predicting the effects of global change (Berner R. et al., 2003; Raymond, P et al., 2013).

The kinetics of silicate weathering are sensitive to temperature, CO_2 emission, and land surface organisation. And previous study by Dessert et al. (2003) and Gudbrandsson et al. (2013) suggests that the order of silicate weathering in nature is similar to Bowen's reaction series of mineral crystallisation (*Olivine> pyx, Ca-rich plagioclase>Amphibole>Biotite, Na-rich plagioclase> K-feldspar>Muscovite>Quartz).* However, there is much debate over their

response strength and evolution. Among all silicate rocks, basalt display a high weathering rate and contributes to a large part of CO_2 consumption flux (Gudbrandsson et al., 2013).

Basalt dissolution is a geochemical process involving the dissolution of basaltic rocks, rich in silicate minerals, in water. Various environmental factors, including the mineralogy and texture of the rock, the temperature, the pH of the water, and the concentration of dissolved CO2 influence the rate of basalt dissolution.

Research has shown that the rate of basalt dissolution is enhanced by increased concentrations of CO_2 in the water (Gudbrandsson et al., 2003). This is because the dissolved CO_2 reacts with water to form carbonic acid, which then reacts with the silicate minerals in the rock and promotes their dissolution. The dissolution of basalt can contribute to the removal of carbon dioxide from the atmosphere, and the resulting dissolved ions can be transported to the oceans and incorporated into the shells of marine organisms. Therefore, it paves a path for us to understand the dissolution and weatherability of other silicate rocks in varying pH condition (acidic to alkaline).

Dissolution of silicate rocks and minerals in the laboratory requires HF which is necessary to break the silica tetrahedra. However, simple equations of natural silicate weathering do not consider HF and typically consider carbonic acid or sulfuric acid. Therefore, the set of experiments that has been carried out in this work is mainly focused on dissolution of geological rock samples with and without the use of HF i.e., HCl and HNO_3. Therefore, understanding the factors that influence the rate and extent of silicate dissolution is crucial for predicting the effects of climate change on Earth's biogeochemical cycles and geologic processes.

1.2 Suite of Samples used:

In this study, a set of acid-leaching experiments have been carried out on five different geological samples- Dunite (DTS-2b, USGS-standard), Basalt (BIR-1a, USGS standard), Deccan Basalt (DVP), Andesite (AGV-2, USGS) and Granite (Bundelkhand craton at Dhala) varying from ultramafic to felsic in composition.

Mineralogy of the samples taken in the experiment:

1. BIR-1a: (Reykjavik Iceland Basalt) is a USGS standard. Rock for this reference material was obtained from the Reykjavik dolerites, a series of lava flows believed to originate from shield volcanoes dating from the youngest interglacial periods. The rock is classified as a coarse-grained olivine tholeiite low in silica content around (45-52%).
2. DTS-2b (Twin sister Dunites): is a USGS standard representing ultramafic composition. Dunite is an olivine-rich endmember of the mantle-derived rocks-peridotite. It has a mineral assemblage of Olivine >90% mineral assemblage, with a minor amount of other minerals such as pyroxene, magnetite, chromite and pyrope.
3. DVP- Deccan Volcanic Province: is a representative of Tholeiitic basalts. The Deccan basalt constitutes mainly Olivine, pyroxenes, plagioclase, and some Fe-Ti rich oxides.
4. AGV-2 (Guano Valley Andesite): It is a USGS standard representative of volcanic rock of intermediate composition. It is collected from the eastern side of Guano Valley in Lake County, Oregon. It predominantly comprises Na-rich Plagioclase and Pyroxene, representing the intermediate type between SiO_2-rich rhyolite and SiO_2-poor basalt.
5. Granite (Bundelkhand Craton at Dhala): This sample was collected from the Dhala Impact Crater in the Bhonti Village in Madhya Pradesh, India. The basement rocks of the impact are predominantly composed of granitoid. This sample is representative of intrusive igneous rocks rich in quartz (>60%) and K-feldspar and plagioclase.

1.3 Aims and Objective of this study:

1. To understand the mobility of cations from silicate rocks comprising silicate minerals with different silicate structures (neso-silicate to tecto-silicate), when reacted with non-HF acids in high pressure and temperature using a High-Pressure Asher (HPA).
2. Geochemical analysis of 'Sadiya" meteorite

Table 1: Suite of samples used in this study.

	Sample Type	**Sample**	**Mineralogy**	**Silica (wt%)**
1	Ultramafic	DTS-2b (Twin Sister Dunite)	Olivine >90%, pyx, magnetite, chromite	~39.4%
2	Mafic	BIR-1a (Icelandic River Basalt)	Coarse grained Olivine tholeiite	~47.96 %
3	Mafic	DVP (Deccan Volcanic Province)	Olivine, pyx, plagioclase, Fe-Ti rich oxides	~52 %
4	Intermediate	AGV-2 (Guano valley Andesite)	Na-rich plagioclase, Pyx	~59.14 %
5	Felsic	Granite (Bundelkhand Craton at Dhala)	Qtz, K-feldspar, plagioclase	~67 %

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Chapter 2
Conventional Method of Digestion:

2.1 Introduction:

The Preparation of samples is a crucial step in geochemical analysis, as accurate and precise data can only be obtained when the sample is completely dissolved. The selection of the best method for sample preparation depends on several factors, such as the nature of the sample matrix, the specific elements being analysed, the analytical technique being used, and the study's intended purpose. All these factors must be considered to ensure the appropriate sample preparation method will lead to reliable results.

Acid Digestion is a commonly and reasonable effectively used method for decomposing geological materials for chemical analysis. The use of acids involved with the majority of digestion are HCl, HNO_3, $HClO_4$, H_2SO_4 and HF, or their combination. The utilisation of HF is a must for many geological samples given their high silica content and its ability to dissolve silicates by the breakdown of silicate bonds (Si-O) by the following reaction (Balram and Subramanyam, 2022):

$$SiO_2 + 6HF \rightarrow H_2SiF_6 + 2H_2O$$

The fluorosilicic acid formed upon further heating breakdown to give gaseous silicon tetrafluoride (SiF_4) and hydrogen fluoride (Dolezal et. al, 1968):

$$H_2SiF_6 \rightarrow SiF_4\,(evp) + 2HF$$

HF is a potent complexing agent; therefore excess ions of fluoride sometimes lead to REE precipitation as fluorides. Therefore, it is usually followed by a next step of acid dissolution with HNO_3 + HCl. HNO_3, a strong oxidising agent, easily dissolves phases like carbonates, sulfides, selenides, tellurides, and arsenides in hot conditions. Whereas HCl is a better acid for the dissolution of Fe-Mn rich phases due to its reducing properties and complexing ability of Cl^-. It helps in dissolving carbonates, phosphates, borates, and sulfates.

2.2 Methods:

2.2.1 Sample preparation:

Beaker Cap digestion for ICP-MS analysis

The samples – DTS-2b, BIR-1a, DVP, AGV-2, DH-2 were taken and processed in a class 100 laminar flow fume hoods in class 10k clean laboratories using established lab-protocols at CEaS, IISc Bangalore.

Approximately, ~25mg of each sample was dissolved in 15mL screw-cap Teflon beakers using a step wise digestion, dissolution, and dilution processes:

1. 1mL of each conc. HF + HNO_3 was added to the sample and heated at 95^0C on hot plate for 24 hours.
2. The dissolved samples' cap was opened on the hot plate at 95^0C and left overnight for evaporation.
3. The evaporated samples were then filled with 1 mL each of HNO_3 and HCl and then left overnight at 95ºC in the same tightly caped beakers.
4. The further dissolved samples in beakers then opened on hot plate at 95ºC and left overnight for evaporation.
5. The acid-digested samples were dissolved using a 2% HNO_3 (120uL) acid and diluted for 6mL centrifuge tubes using 18.2 milliQ water.
6. The dilution factor was calculated as DF = final wt. of the sample solution (mg)/initial weight of sample taken(mg). Generally, DF is kept at ~4000 for trace element analysis and ~10000 for major element analysis.
7. The United States Geological Survey (USGS) rock standards BHVO-2 (Hawaiian basalt), BCR-2 (Columbia River Basalt), AGV-2 (andesite) and BIR-1a (Reykjavik Iceland Basalt) were also prepared in a similar way.
8. The USGS rock standards were diluted 4000 times for TE analysis and 10000 times for ME.

2.2.2 Element composition analysis in ICP-MS:

The element composition of the samples was determined using Thermo-Scientific X Series Inductively, Coupled plasma Mass Spectrometer (ICPMS). Total of 20 Sample solutions of 10 leached solutions and 10 residue solution from 2 set of experiments, 4 rock standards,1 Blank were introduced from CETAC ASX-520 autosampler through a nebuliser into the spray chamber.

The analytes selected were REEs (La, Ce, Pr, Nd, Sm, Eu, Gd, Tb, Dy, Ho, Er, Tm, Yb, Lu), Trace elements Li, Sc, V, Cr, Co, Ni, Cu, Zn, Ga, Rb, Sr, Y, Nb, Ba, Y, Ba, Hf, Ta, Pb, Th, U, Cd, Pb and Major elements: Na, Mg, Al, Ti, Fe, Mn.

The $^{156}CeO/^{140}Ce$ ratio was kept below 0.05 (5%) for to minimise oxide-related interferences.

^{7}Li, ^{115}In and ^{238}U were selected to scan the mass range for the detector using Tune-A solution.

A three-point calibration method was used to calculate the concentrations using the detector counts of the known USGS standards. All detected counts were corrected for contribution from acid blank (2% HNO_3 solution). USGS rock standards BHVO-2, BCR-2 and AGV-2 were used as known calibration standards, whereas BIR-1a was run as unknown. The samples were run in 2 blocks and the accuracy of the data was estimated by repeated measurements of BIR-1a and AGV-2 as unknown.

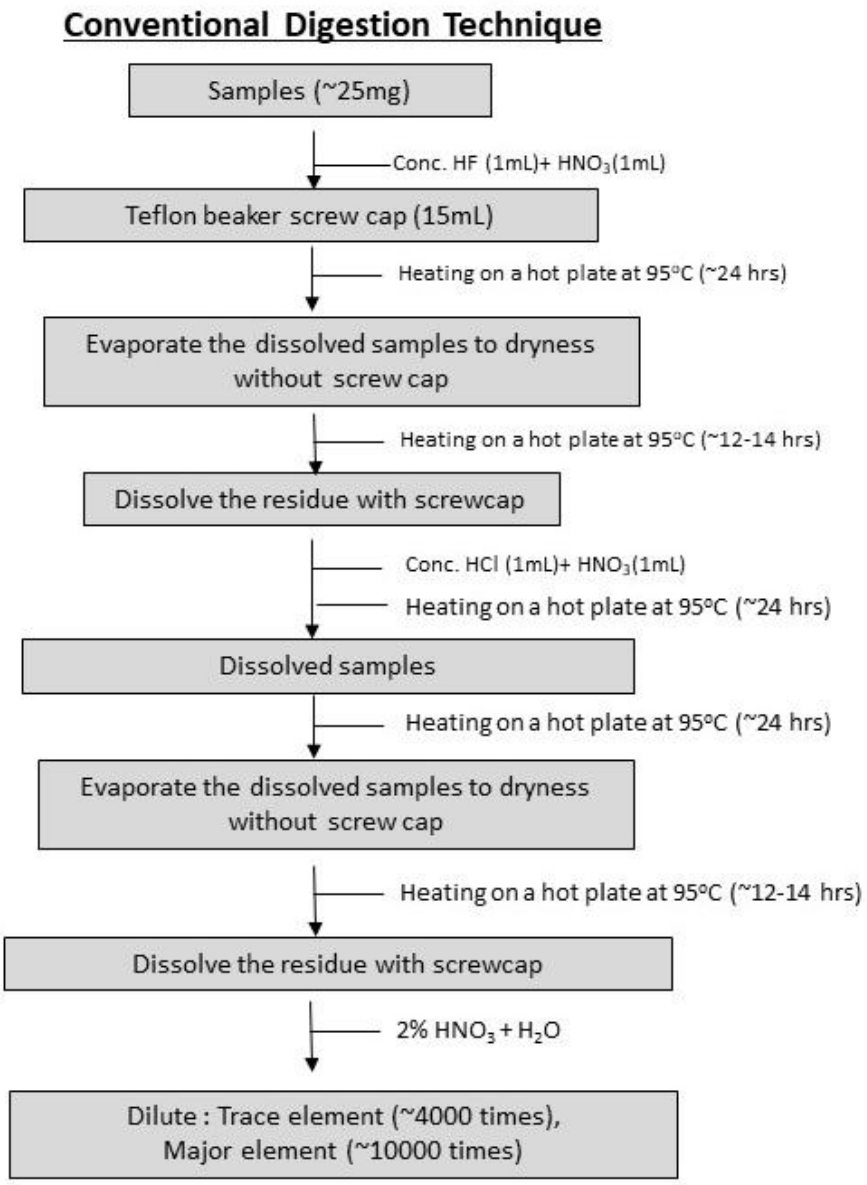


Figure 2.1: Flowchart depicting methods followed for conventional way of beaker-cap digestion technique.

Chapter 3
High Pressure Asher (HPA) Digestion

3.1 Introduction:

High Pressure Asher (HPA) digestion techniques is a popular method for sample preparation in geochemistry due to their ability to dissolve a wide range of sample types while minimising contamination. HPA digestion involves heating samples under high pressure with strong acids, resulting in complete digestion of the sample matrix and breakdown of mineral phases. This technique has been applied in various geochemical studies, including the analysis of trace elements, isotopes, and organic compounds. Since, 1980s, the HPA (High-Pressure Asher) has been a unique sample preparation system because of its ability to quickly and fully decompose organic materials. The process of decomposition took place in enclosed quartz vessels at the desired reaction temperatures, up to 320°C, under a pressure of 100 bar. Compared to other sample preparation techniques of that time, the HPA was one of the most advanced instruments available. Since then, several methods have been developed using HPA technology, which can handle a wide range of sample sizes and challenging matrices.

The basic principle for HPA-S successful wet chemical digestion depends upon three major factors:

1. Amount of sample used.
2. The appropriate volume of Acid used.
3. Temperature

In order to reach temperatures beyond the boiling points of acids, it is necessary to work within a closed vessel under pressure. The maximum temperature and sample size that can be used depend on the pressure capability of the digestion vessel. The most suitable materials for these vessels include quartz glass, certain fluoropolymers such as PTFE and PFA, and glassy carbon. The use of gastight, closed vessels helps minimise the risk of contamination or loss of analytes, and also requires smaller amounts of reagents, ultimately reducing the possibility of systematic errors.

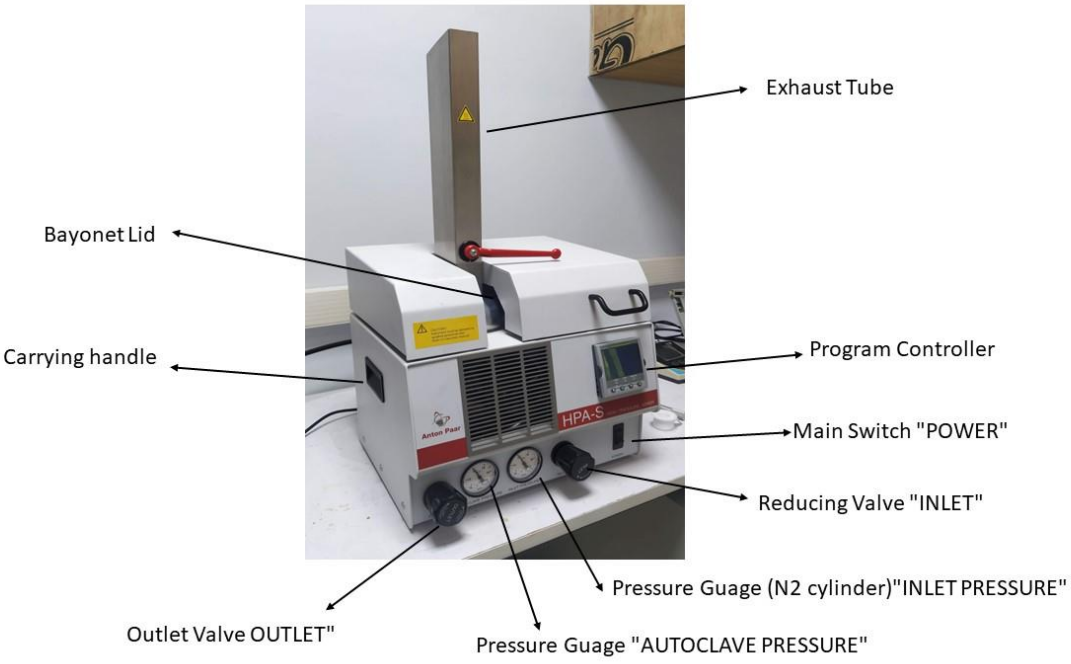


Figure 3.1: High Pressure Asher -Anton Paar, CEas, IISc

3.2 High Pressure Asher-Parts and their functioning:

3.2.1 Reaction Vessels:

The HPA-S may require the use of vessels made from either quartz glass or glassy carbon. These materials are known for their ability to withstand high temperatures and their purity. They are also gastight and pressure-tight and can be easily handled without requiring specialised tools (fig 3.2 a)

3.2.2 Sealing of the quartz vessel:

Lids made from the same material as the vessels, i.e., quartz and a strip of analytically-pure PTFE tape complete the vessel assembly. A wrapping aid has been developed for the HPA-S

system, making vessel sealing very reproducible. The vessel assembly for the HPA system consists of vessel lids and specially produced low-blank PTFE tape.

The sealing process involves placing a strip of low-blank PTFE tape over the vessel opening, followed by placing a quartz lid on top. The lid is held in place with a weighted plunger on the wrapping aid tool while two or three coils of PTFE tape are wrapped around the vessel neck and lid (fig 3.2 b). This method is cost-effective, safe, and provides a disposable seal.

3.2.3 Heating:

The system is heated electrically to the desired temperature, up to a maximum of 320ºC. The chemical reaction during digestion can generate pressure in the vessels, which is equal to the vapor pressure of the acid(s) plus the partial pressures of the reaction gases. To prevent the vessels from rupturing, the nitrogen pressure of the autoclave keeps them hermetically sealed, even at the highest temperature, for the required amount of time (fig 3.2 f).

3.2.4 Heating Block and Autoclave:

The heating blocks used in the HPA-S system are designed to closely and efficiently transfer heat to the vessels, which are tightly sealed to ensure good mixing of the sample and reagent by convection. Vertical temperature gradients within the vessel further aid in mixing of the sample with the acid. Once the vessels are sealed in the heating block, they are placed in the autoclave of the HPA system and quickly sealed using a pressure-tight cover with a quick-fit bayonet seal (fig 3.2 d). This design is both user-friendly and safe, as it minimises the risk of user error and spontaneous reactions.

While the sample is being digested, the vessel is positioned inside the heating block, which is located inside a heatable pressurised container (autoclave). The autoclave is filled with nitrogen up to 100 bar, which causes pressure to build up on the outside of the vessel and lid, resulting in a secure seal that protects the sample from potential sources of contamination during decomposition.

The autoclave undergoes electrical heating at a power input of 1600 W, which enables a temperature increase from ambient to 250°C within 20 minutes. The attainment of 300°C necessitates approximately 40 minutes, with the ultimate temperature limit being 320°C. The

HPA's heating blocks are arranged concentrically within the circular pressure chamber, providing even radial temperature distribution and ensuring uniform digestion. The crucial prerequisite for achieving analytical reproducibility is the highest degree of uniformity in temperature across vessels and runs.

3.2.5 Temperature Programs:

The integrated program controller allows for the storage of multiple temperature programs, each with up to 16 segments including ramps and holding times. Empirical evidence has demonstrated that three to four uncomplicated programs are typically satisfactory for achieving optimal digestion of any sample type. The selection of heating and maximum temperatures is dependent upon the sample's reaction characteristics, weight, and the vessel's volume. However, for this study, only two experiments were run – first at 280°C and second at 300°C.



Figure 3.2 (a) Quartz reaction vessel (b)Sealing of quartz vessel using PTFE tape (c) Heating block with capped quartz vessel (d) Running HPA (e) Temperature program section (f) Heating block placed inside Autoclave.

3.3 Methods:

The samples were processed in a class 100 laminar flow fume hoods in class 10k clean laboratories using established lab-protocols at CEaS, IISc Bangalore.

1. Approximately ~50 mg of each sample DTS-2b, BIR-1a, DVP, AGV-2, Dhala are taken in 15 mL quartz vessel.
2. 2.5mL (HNO_3) + 1mL (HCl) is added. - (Reverse aqua regia) And sealed carefully using PTFE tape.
3. Two sets of experiments were carried out: first is run at 280^0C for 3 hrs and the second with increased temperature and timing of digestion at 300°C for 5 hrs.
4. The leached solution for both experiments is then transferred carefully into 15 mL screw-cap Teflon Beakers whose empty weights were previously measured. The leached solution in beakers were left overnight for evaporation on a hot plate at 95°C. The evaporated solution leaves behind the salt which is then measured for their weight.
5. Similarly, the residue left behind in the quartz vessel is retrieved using 18.2 milli-Q water and transferred carefully into 15 mL Teflon beakers whose weights were previously taken. The residue within water solution is left overnight for evaporation on hot plate at 95°C. The measured weights of leachates and residues from experiments I and II are given in table 2.
6. The leachate is then diluted with 2% HNO_3 and 18.2 milli-Q water for ICP-MS analysis.
7. The residue is further treated with HF followed by HNO_3 and HCl to break-free silicate bonds for elemental analysis in ICP-MS.

Procedure followed during High Pressure Asher digestion is explained in flowchart (fig 3.3)

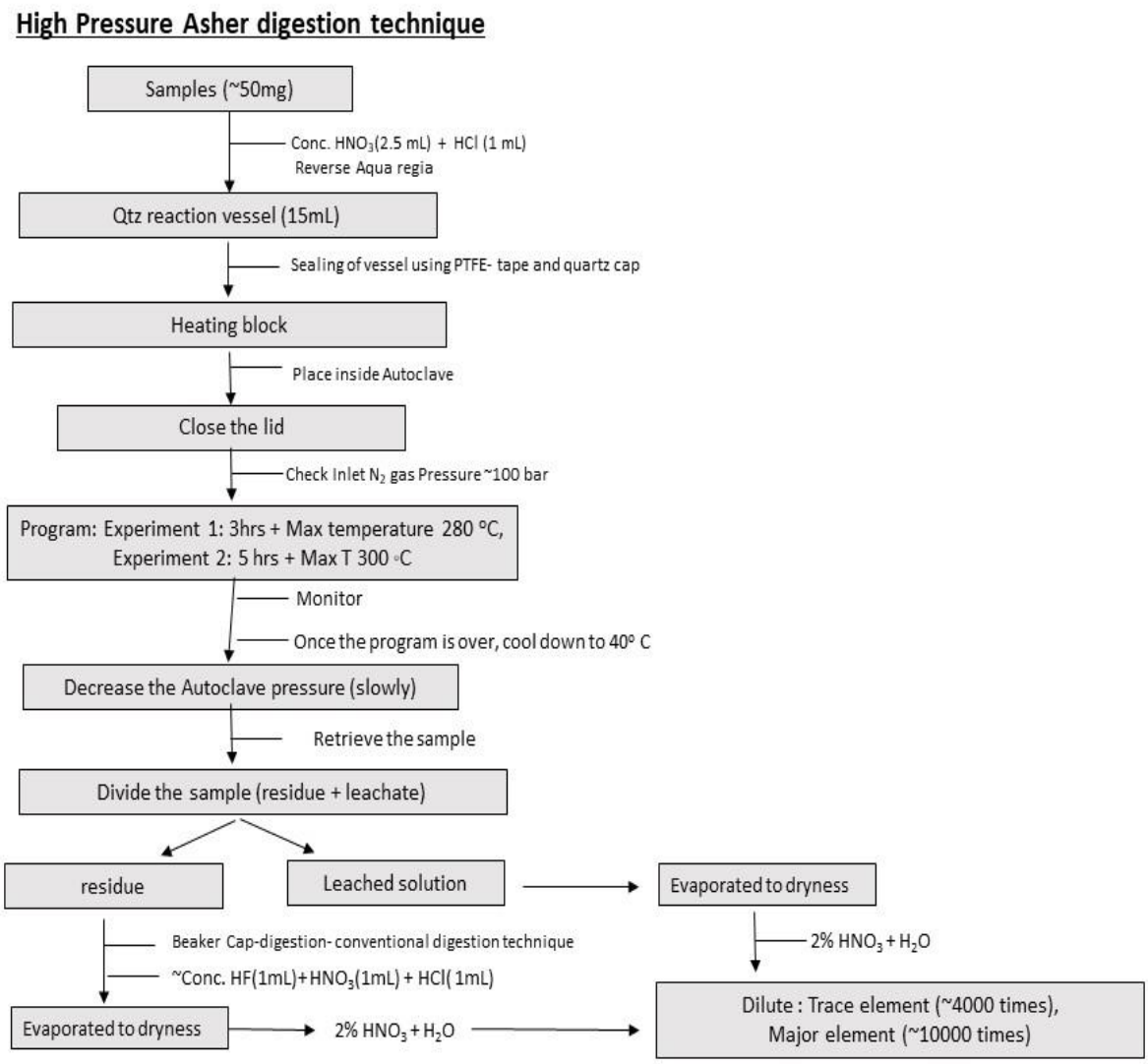


Fig 3.3: Flowchart of methods followed for digestion of samples in High-Pressure Asher

Table 2: Weight measured for Leachate and Residue in Experiments I and II

Experiment - I				Experiment -II		
No.	sample	Exp-I-Leached solution wt (mg)	Exp-I-Residue wt (mg)	Exp-II-Leached solution wt (mg)	Exp-II-Residue wt (mg)	Average wt Residue (mg)
1	DTS-2b	39.8	10.2	40.1	9.9	10.05
2	BIR-1a	24.2	25.8	35.8	14.2	20
3	DVP	20.9	29.1	10.4	39.6	34.35
4	AGV-2	14.8	35.1	17.7	33.2	34.15
5	DH-26	10.4	25	9.9	27.5	26.25

3.4 Results and discussion:

3.4.1 Rare Earth Elements Spider Diagram

Evidently from the diagram, light rare earth elements (La, Ce, Pr, Nd, Sm, Eu, Gd) are enriched in all the samples compared to heavy rare earth elements (Tb, Dy, Ho, Er, Tm, Yb, Lu), as LREEs are more mobile than HREEs. The REEs concentration in the felsic rocks like Granite, Andesite and Deccan Basalt are significantly higher than the mafic and ultramafic rocks like Icelandic-river basalt and dunite which are in accordance with the bulk values (Fig 3.4 and 3.5). The Eu anomaly observed in the Fig 3.4 is related to the Ca concentration, which is further explained in the following section.

3.4.2 Partition Coefficient (K_D):

While the Deccan basalt, andesite and granite are showing almost ten folds enrichment of the REEs concentration in the leachate (leached solution), compared to the residue, the REEs concentration in the Dunite and the Icelandic-river basalts both in residue and leachate solution are almost indistinguishable. This can be explained by the partitioning coefficient (K_D) which for an element is the ratio of the concentration of that element in the solid to that in the liquid. It can be seen in Fig 3.6 that the K_D values of the rocks Granite, Andesite, Deccan basalts are almost 10 times lesser than 1 while for the Dunite and the Icelandic basalt it is close to 1.

The anomalous behaviour of the Eu can be explained by similarities with Ca in terms of partitioning behaviour between residue and leachate (Fig.3.5). Ca shares similar charge and ionic radius with Eu and hence can be replaced efficiently. Hence the rocks like Dunite, BIR, showing more than 1 K_D value for Ca (Fig 3.6) are also showing Eu enrichment in residue compared to leachate while the Granite, Andesite and Deccan Basalts are showing opposite behaviour.

The elements like Li (Group I), Mg (Group-II) are showing K_D values equal to or less than 1, exhibiting the natural tendency of these elements to get leached into the solution under acidic condition, similar to continental weathering phenomenon. On the contrary, Na, K, Rb, Sr are showing K_D values equal or more than 1, demonstrating their nature to remain in the solid except for deccan basalt. Ca is showing similar behavior as Eu, as discussed earlier.

3.4.3 Relation between Silica Content and Weight of the Residue:

In terms of SiO_2 wt%, Andesite and Granite are classified as intermediate and felsic rocks, respectively as they have dominantly quartz and feldspar in their mineralogy. Deccan represents continental flood basalts, classified as quartz tholeiite which dominantly has pyroxene, feldspar, and quartz in their mineralogy along with minor or no Olivine. One the contrary, dunite (DTS) is Olivine dominated monomineralic rock and Icelandic-basalt (BIR) is classified as olivine tholeiite with minor or no quartz. Since the digestion technique used here does not include any Hydrofluoric acid (HF) which is essential to break the strong silicate bonds, e.g., the tectosilicate bonds in quartz and feldspar. Hence, the rocks dominantly containing high wt % of SiO_2 are showing high ratio of residue upon total weight than those with the least SiO_2 (wt 5%) (fig 3.7) This also represents the stability order of the minerals, increasing from ultramafic to felsic rock types in nature.

3.4.4 Individual REEs concentrations in residue, leachate, and true values (Fig 3.8):

Fig 3.8 (A-E) demonstrates the chondrite normalised REEs concentrations for all the rocks for both residue, leachate and bulk values. The spider patterns are similar for all the rocks where the plot of the bulk values lie between the residue and the leachate, except for Icelandic-Basalt (BIR-1a). It is showing erroneous data which could be because of the presence of moisture trapped as a function of measured weight of the leachate and residue while taking into consideration for the measurement of dilution factor in ICP-MS analysis.

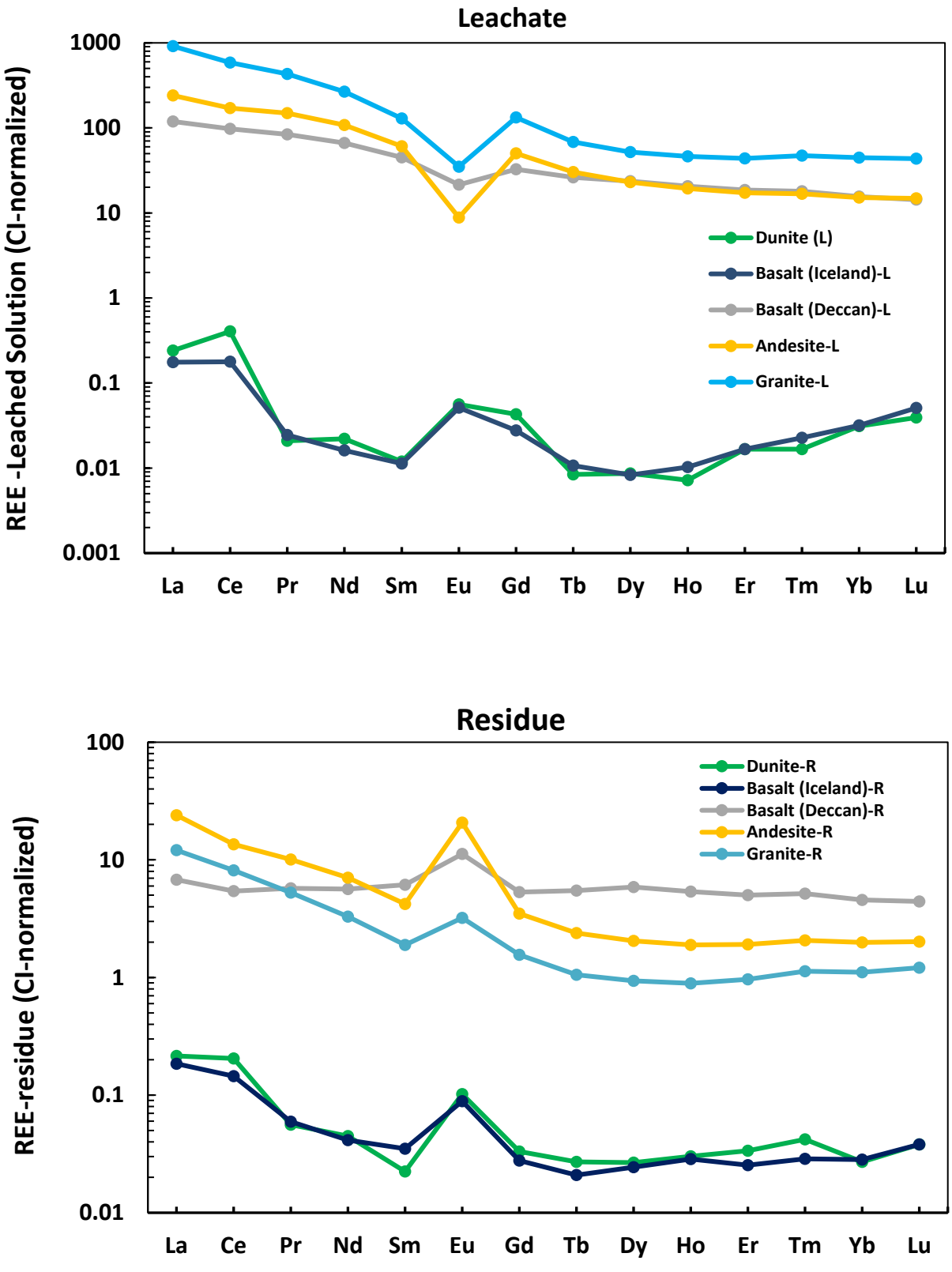


Fig 3.4: Chondrite normalized REEs concentrations (in logarithmic scale) for all the rocks both in residue and leachate.

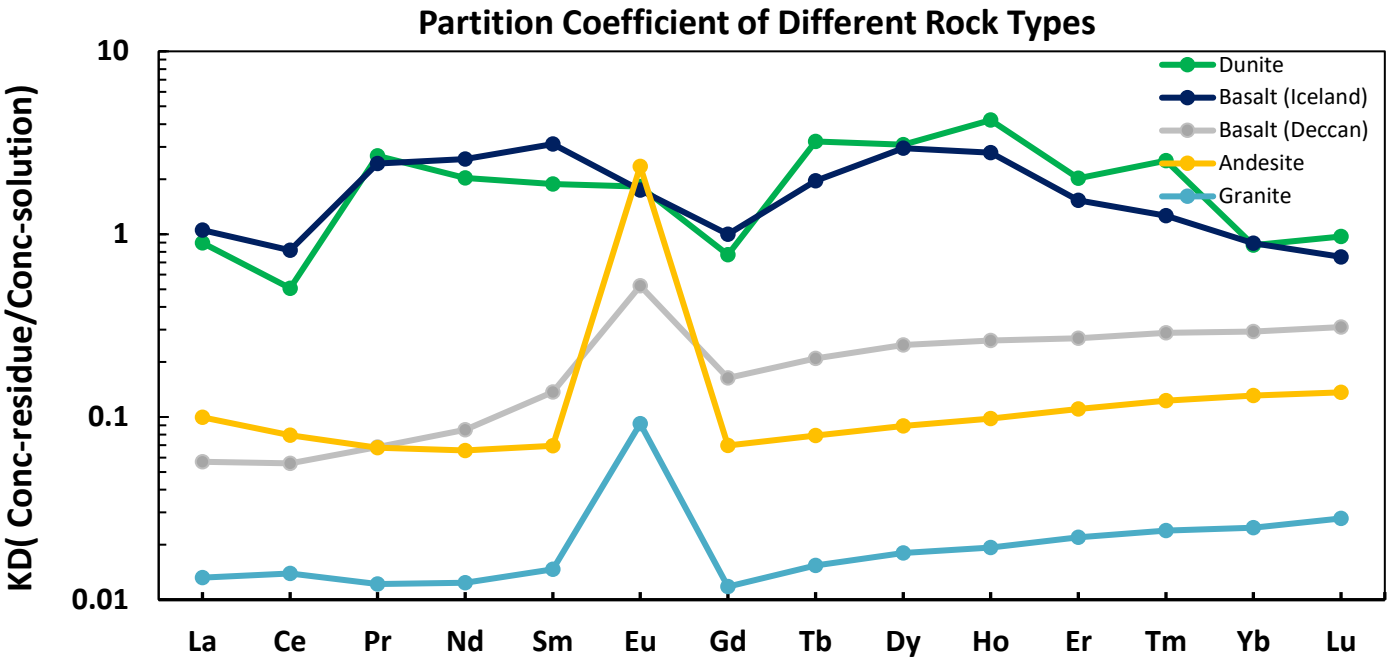


Fig 3.5: Partition coefficient of REEs between residue and leachate for all the rocks.

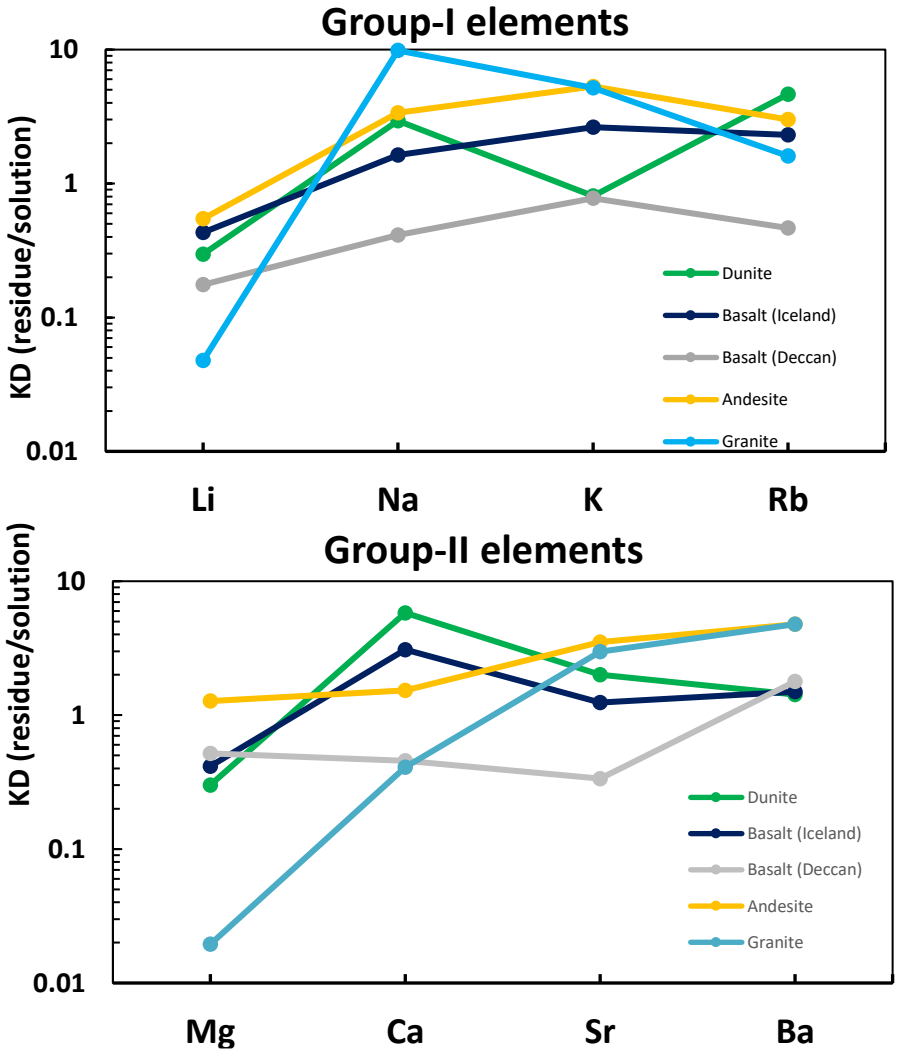


Fig 3.6: Partition coefficient of Group I-II elements between residue and leachate.

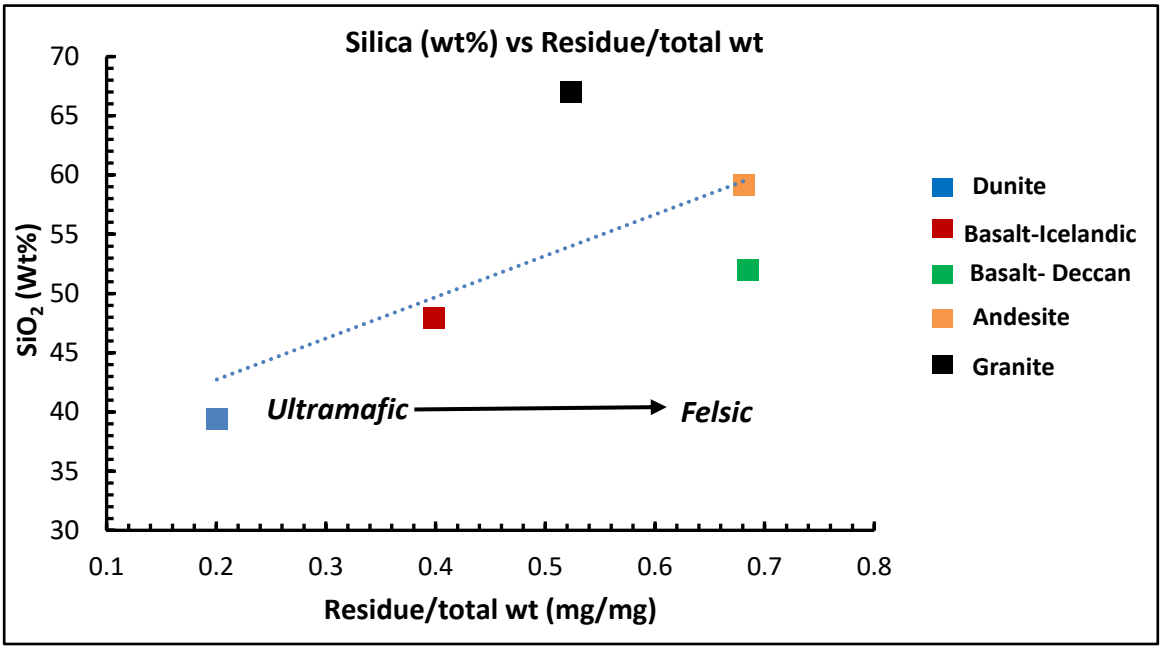
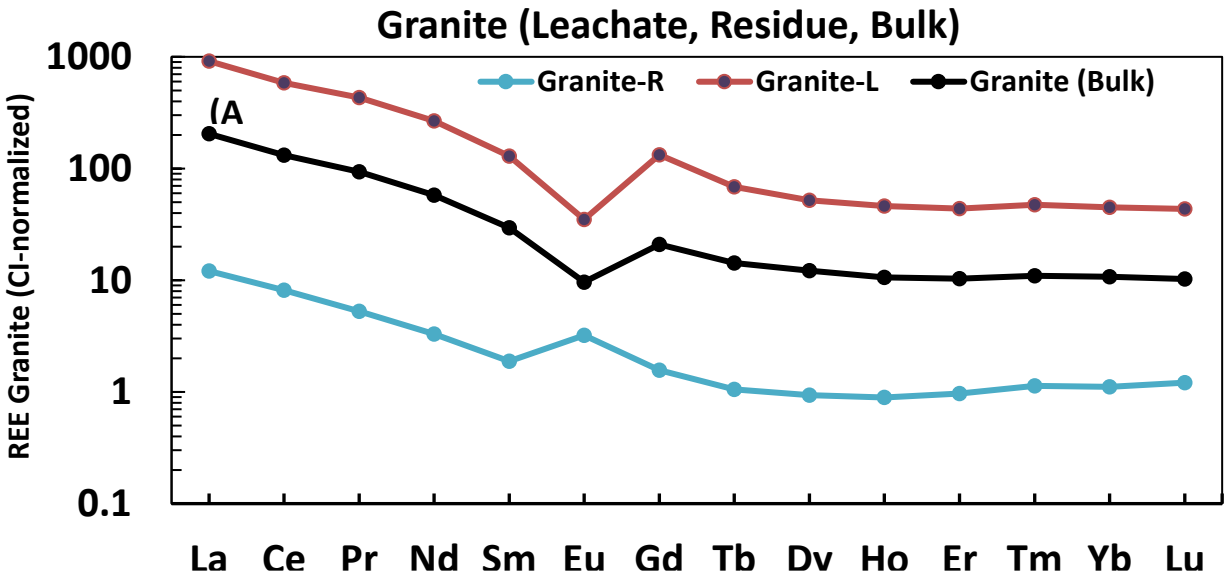
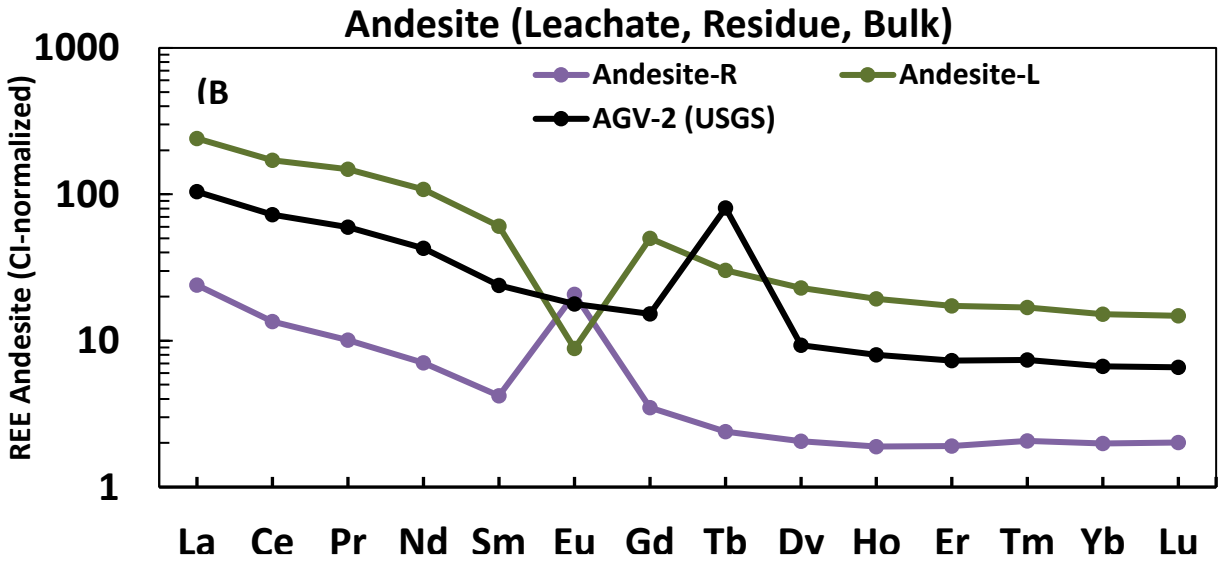


Fig 3.7: Relation between Silica wt% and relative weight of residue.





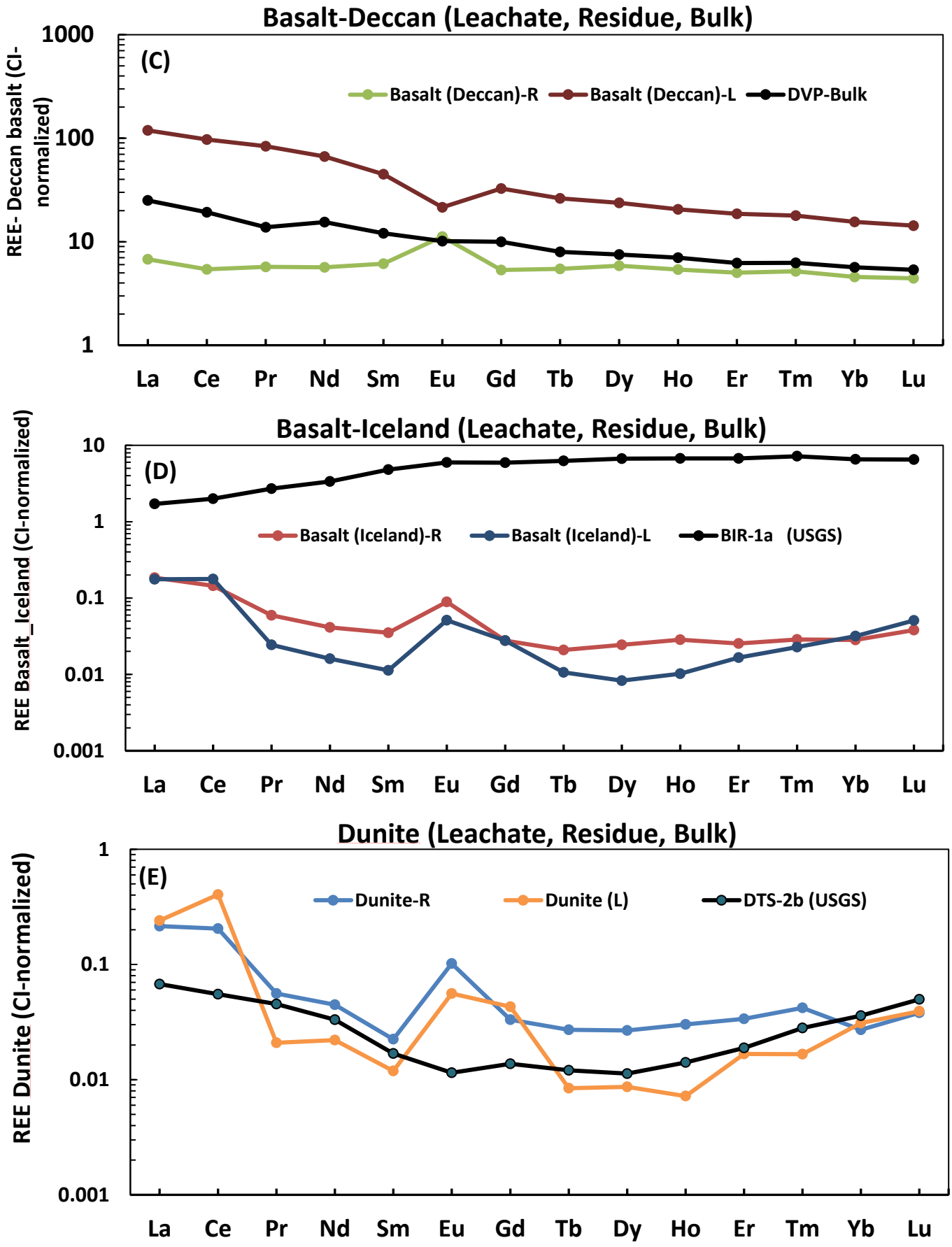


Fig 3.8 (A-E): REE concentration (CI normalised) of Leachate, Residue, and Bulk of each rock types.

3.5 Conclusion:

A new digestion method was used in this study that excludes hydrofluoric acid (HF), which is essential to break the strong silicate bonds. The residue weight shows a positive correlation with SiO_2 wt%, consistent with the mineral stability order from felsic to ultramafic. The LREEs show relative enrichment in all the rock types, compared to HREEs, as they are more mobile. The partitioning of the REEs between residue and leachate is much more significant in the felsic rocks than mafic and ultramafic rocks, which is evident from the partition coefficient close to 1 for Icelandic-basalt and Dunite while it is ten folds lower for Andesite, Deccan-basalts and Granite. The Eu anomalies in these rocks are in accordance with Ca anomalies, demonstrating the geochemical behavioral similarities between these two elements.

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Chapter 4
Geochemical Analysis of 'Sadiya' Meteorite

4.1 Introduction:

Ordinary Chondrites are undifferentiated and include the H, L, and LL groups which stand for High-Iron, Low-Iron, and Low-Iron-Low-Metal, respectively. The H, L, and LL groups make up typical chondrites. More than 85% of observed falls are O chondrites, which make up the majority of the materials in our meteorite collections. They are distinguished by an extreme quantity of large (millimetre-sized) chondrules with a range of textures and mineral makeups. Compared to Carbonaceous chondrites, matrix abundances (10–15 vol%) are typically lower; CAIs and AOAs are rare. Ordinary chondrites exhibit various petrologic types, from 3 to 6, with a few less-altered (petrologic types 3.0-3.1) members.

Aqueous alteration can be seen in some of the least metamorphosed O chondrites, such as Semarkona and Bishunpur (Hutchison et al., 1987; Alexander et al., 1989; Sears and Weeks, 1991; Sears et al., 1980, 1991). According to Hutchison et al. (1987), the matrix and occasionally the chondrule mesostases were the major sites of hydrous alteration. According to Brearley and Jones (1998), Bishunpur appears to have undergone less change than Semarkona. The degree of change varies. Their O-isotopic compositions map above the terrestrial fractionation line and their whole rock Mg-normalised refractory lithophile abundances are 1 CI, clearly separating them from the C chondrites. The O-isotopic compositions and general petrologic traits of the H, L, and LL chondrites show that they are related to one another and belong to the same clan.

Numerous intriguing interpretations had been developed in response to the existing OC data. Some were based on nonsystematic and poorly understood compositional differences, while others were based on systematic trends reflecting regional differences in the formation of planetesimals. The three groups are merely fragments of an incompletely sampled continuous fractionation sequence, according to Tandon and Wasson (1968) and Moller et al. (1971) who also interpreted systematic differences in OC degree of oxidation and siderophile quantity. The OC formational zone would have homogenous nebular metal and silicate components, according

to a model assessed by Larimer and Anders in 1970. Interelement fractionations among siderophiles were discovered by Moller et al. (1971), Chou et al. (1973), and Rambaldi et al. (1979), indicating the presence of multiple metal components in the nebula.

4.2 Sadiya Meteorite and its Classification:

Sadiya is an observed fall which occurred on June 5, 2017 with a weight of 3 kg and density of 3.4 g/cm^3 (Saikia et al., 2017). From the viewpoint of a cosmochemist, the main purpose of a chondrite classification is to group together chondrites that appear to have originated in the same parent body. In this study, an attempt has been made to classify the group in which this ordinary chondrite lies. The H, L, and LL groups are resolved using various parameters. Metal abundances and chondrule size show minor, systematic variations among the ordinary chondrite groupings. It is also possible to discriminate between H, L, and LL using siderophile element abundances (Kallemeyn et al., 1989; Sears et al., 1991). This latter method will be used as only bulk composition data has been measured in this study.

The Mg and CI-normalised group data for Ni, Co, and Fe are shown in Fig. 4.1. From Kallemeyn et al., 1989, it has been understood that the Mg-normalised abundances of these elements can be used to distinguish between the three groups of ordinary chondrites. These elements were largely in nebular siderophile components during siderophile-lithophile fractionation. Because their condensation temperatures are virtually identical to those of Fe, it is convenient to also define Ni and Co to be common elements. Although an appreciable fraction of Fe was oxidised in the nebula, Co remained strongly siderophile and almost entirely present in metallic components. The surprising discovery made by Fulton and Rhodes (1984), and confirmed by Kallemeyn et al. (1989), is that the Co/Ni ratio decreases (by 5%) through the H-L-LL sequence.

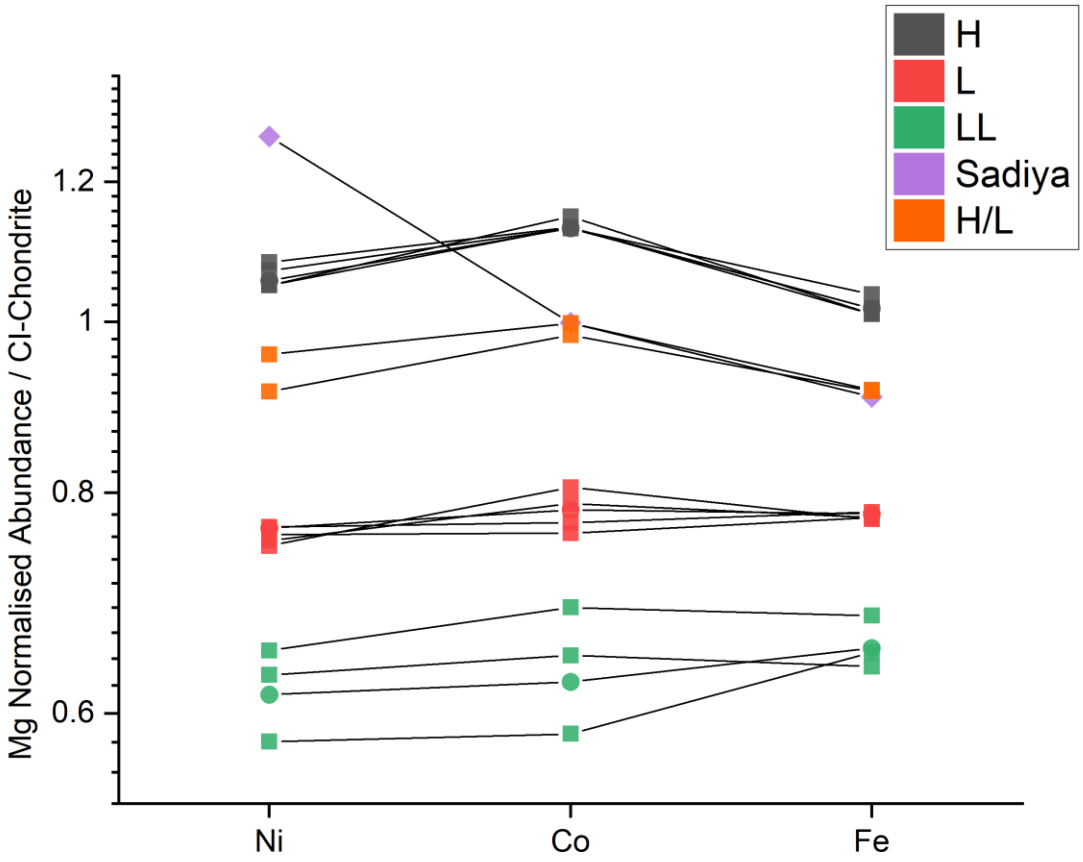


Fig 4.1. Ni, Co, and Fe abundances of different group of Ordinary Chondrites normalised to Mg and also normalised to CI-Chondrites. The literature values have been taken from Kallemeyn et al., 1989. Bremervörde and Tieschitz are the two-sample characterised as H/L chondrites by Kallemeyn et al., 1989, as they lie between H and L group. The square shapes are the data points for individual samples whereas the circles are mean of their respective groups.

As can be seen in Fig. 4.1, The Mg-normalised abundances of Co and Fe suggests that Sadiya can be classified as H/L, although Ni's concentration is significantly high compared to any group of ordinary chondrites. 56 meteorites have been classified as H/L group as per the Meteoritical Bulletin, out of which Bremervörde and Tieschitz are the two of the four falls.

Some moderately volatile elements are also useful taxonomic parameters because they show relatively constant abundances or concentrations within groups but relatively large intergroup variations. Fig. 4.2 displays a plot of K vs Na for different groups of carbonaceous, enstatite, and ordinary chondrites.

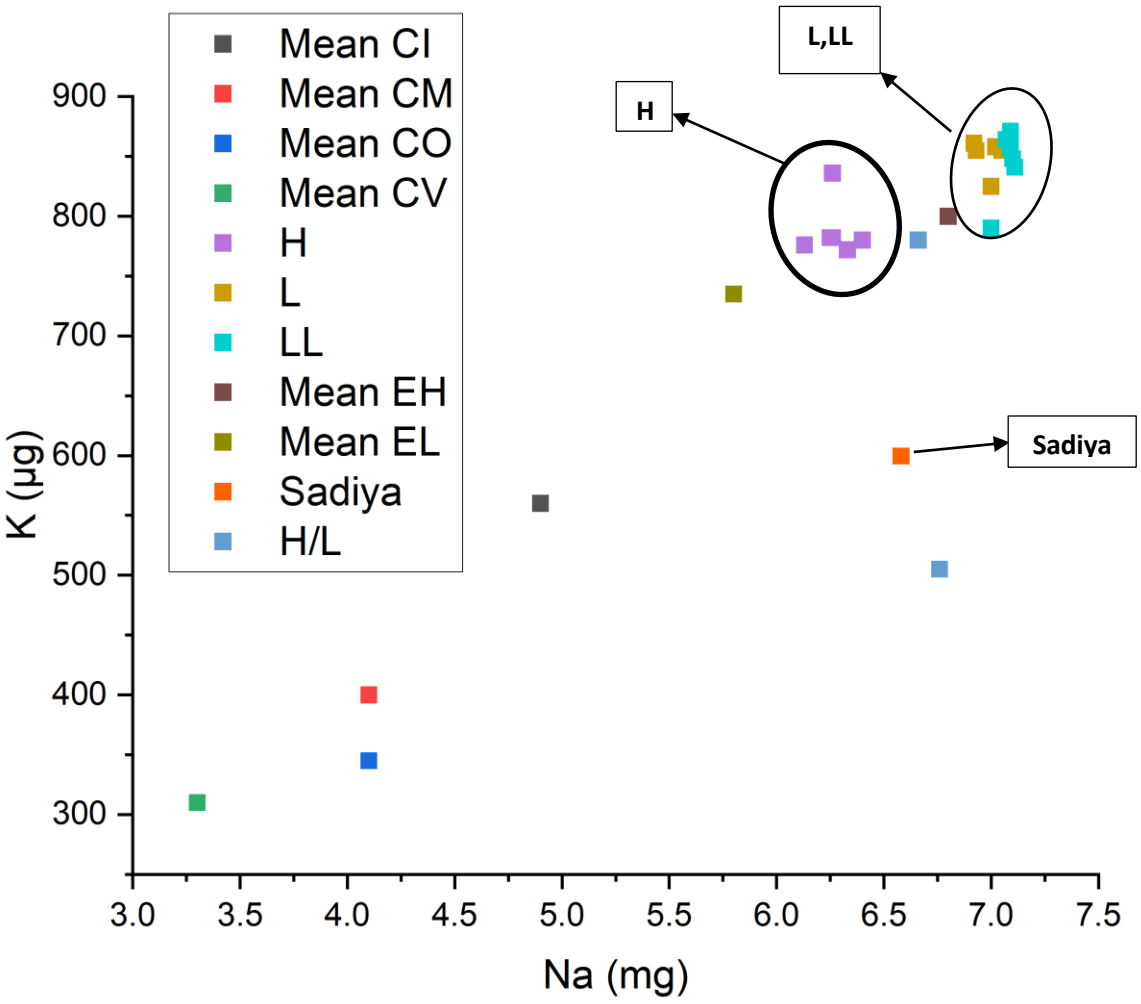


Fig 4.2. K/Na plot for different groups of Chondrites. Note that the groups are consistent, whereas the H/L chondrites (Sadiya, Bremervörde and Tieschitz) are quite dispersed. The mean values of carbonaceous and enstatite chondrites are taken from Wasson and Kallemeyen (1998), and the data for individual ordinary chondrites is taken from Kallemeyen et al., 1989

On the other hand, the petrographic type of ordinary chondrites is extremely difficult to distinguish without petrographic and textural analyses, just by using geochemical data. Kallemeyn et al., (1989) attempted by plotting Mg-normalised abundances for H, L, and LL groups (Fig.4.3.).

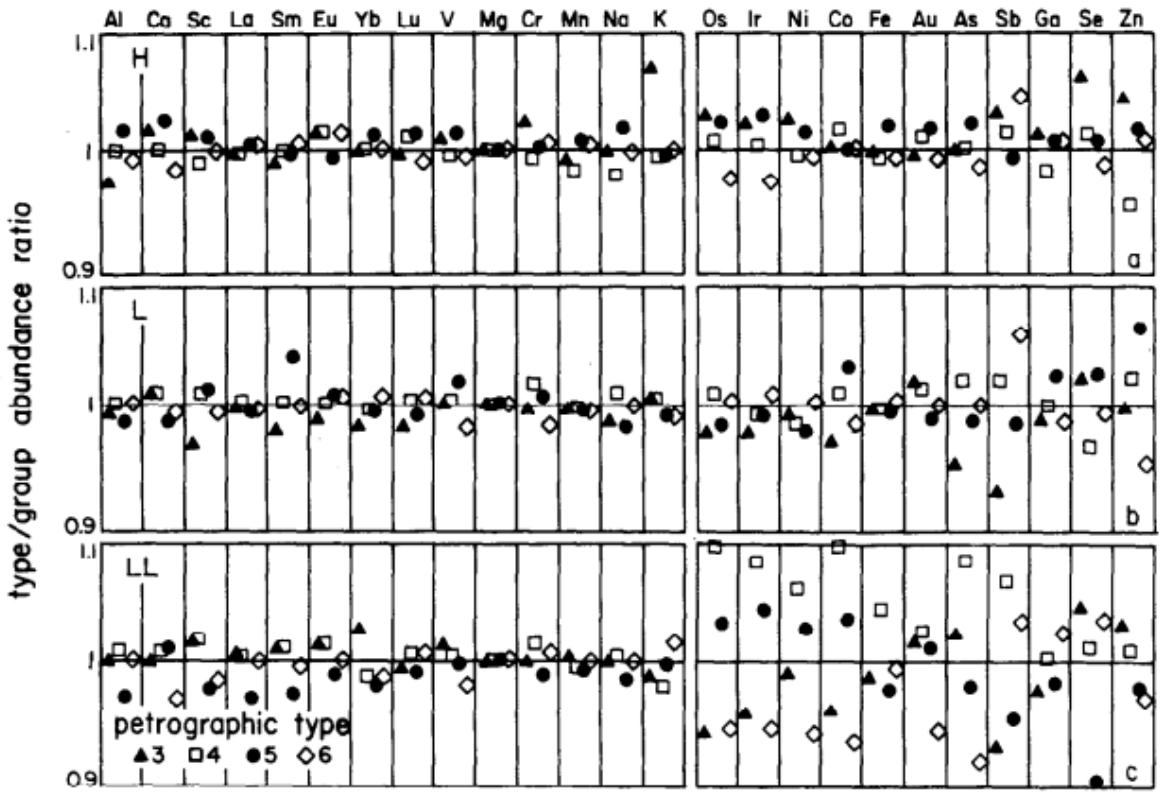


Fig 4.3. No significant differences in abundance patterns are observed among the four petrographic types of the H, L, and LL groups (Taken from Kallemeyn et al.,1989). Therefore, it is not possible to determine the petrographic type of Sadiya by just using the bulk composition.

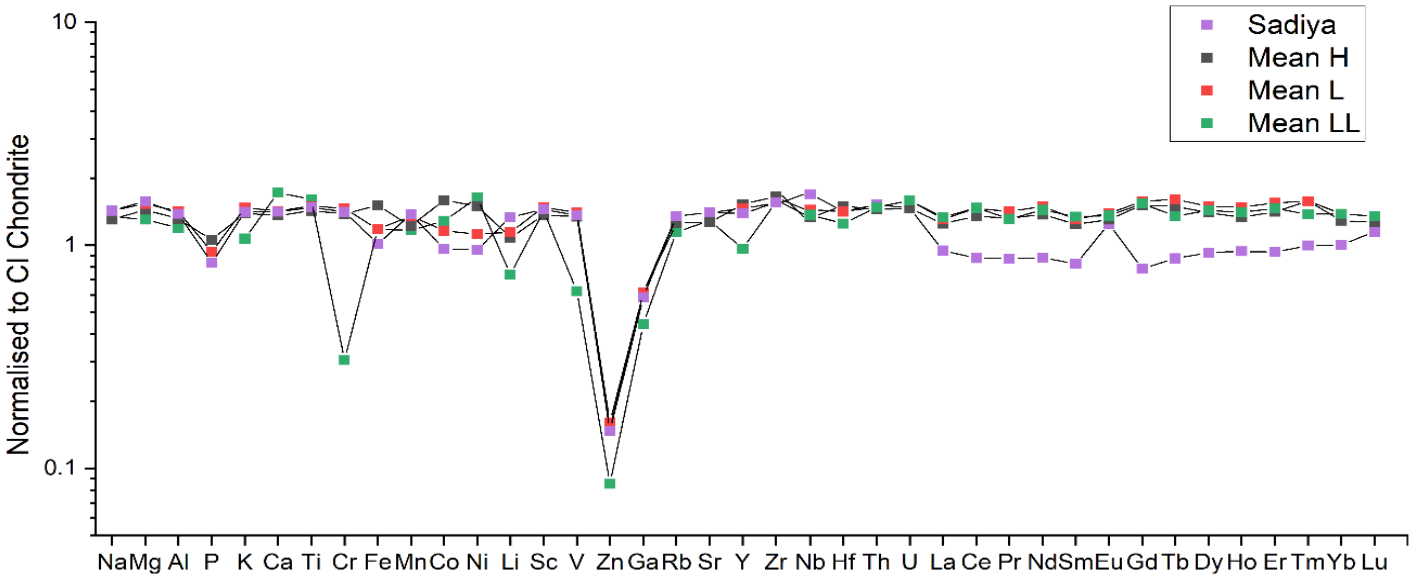


Fig 4.4 CI Normalised Trace Element Abundances of Sadiya. The Mean H, Mean L, Mean LL patterns have been displayed for comparison. The Mean H, L, and LL chondrites values have been taken from Wasson and Kallemeyn, 1988.

4.3 Conclusions:

Ordinary chondrites, being undifferentiated, are crucial in understanding the nebular as well as asteroidal processes. In this study, Sadiya was characterised as H/L chondrite based on moderately siderophile elements. The volatile elements also show that Sadiya does not fit within ordinary chondrites' three groups (H, L, LL). Therefore, the possibility of a fractionating range of ordinary chondrites should be explored to accommodate the several meteorites currently grouped as H/L, L/LL, etc. Moreover, determining petrographic type is important to understand a chondrite's thermal metamorphism or aqueous alteration history. However, it cannot be determined without petrographic and textural studies.

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Appendix

Table 1: REEs concentration of leached(ss) and residue(resi)

SAMPLES	La	Ce	Pr	Nd	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
	ppb	ppb	ppb	ppb	ppb	ppb	ppb	ppb	ppb	ppb	ppb	ppb	ppb	ppb
DTS (2) SS	88.31	386.7	2.866	15.71	2.75	4.872	13.15	0.488	3.291	0.612	4.155	0.593	7.689	1.494
BIR(2) SS	64.53	170.1	3.347	11.43	2.608	4.444	8.487	0.619	3.161	0.872	4.141	0.81	7.868	1.933
DVP(2) SS	43690	93100	11490	47290	10360	1871	9974	1522	9034	1752	4640	638	3864	545
AGV(2) SS	88360	163500	20360	76630	14000	768.4	15340	1754	8750	1647	4311	599.1	3753	563.8
DH(2) SS	335500	559800	59150	189100	29760	3047	40490	3970	19780	3926	10930	1684	11110	1656
DTS (2) Resi	78.98	195.7	7.668	31.84	5.176	8.871	10.13	1.57	10.16	2.572	8.396	1.495	6.703	1.448
BIR (2) Resi	67.83	138.9	8.127	29.43	8.107	7.74	8.487	1.212	9.315	2.431	6.334	1.023	7.016	1.452
DVP (2) Resi	2482	5189	784.4	4022	1416	975.4	1631	317.8	2234	458.2	1251	184.1	1134	168.8
AGV (2) Resi	8805	12960	1381	5024	972.7	1805	1070	138.7	781.6	161	475.5	73.6	491.5	76.85
DH (2) Resi	4433	7787	720.6	2341	435.7	279.8	477.6	61.06	356.2	75.89	240	40.21	275.3	46.12

Table 2: Major Elements concentration of leached(ss) and residue(resi)

Sample	Na	Mg	Al	Ca	Ti	Fe	Mn
	ppm	ppm	ppm	ppm	ppm	ppm	ppm
BIR-2-ss	54.53	318600	2007	451.6	10030	64580	846
DTS-2-ss	32.83	305600	1804	348.3	8977	61490	927.3
DVP-2-ss	28750	69170	168600	119800	955.9	190900	2091
DH-2-ss	3957	13660	41960	24000	858.9	53540	1250
AGV-2-ss	11650	8997	23090	27020	1638	95150	1308
BIR-2-RESI	89.09	132500	1329	1393	4424	26580	394.8
DTS-2-resi	96.27	92000	1100	2023	3398	18690	278.8
DVP-2-resi	11850	35840	38630	54740	7532	43330	898.3
DH-2-resi	38980	266	111300	9815	1598	1301	35.43
AGV-2-resi	39390	11460	119500	41310	7238	22380	514.6

Table 3: Partition Coefficient (K_D) values of REEs

Partition Coefficient	**La**	**Ce**	**Pr**	**Nd**	**Sm**	**Eu**	**Gd**	**Tb**	**Dy**	**Ho**	**Er**	**Tm**	**Yb**	**Lu**
Dunite (DTS-2b)	0.894 349	0.506 077	2.675 506	2.026 735	1.882 182	1.820 813	0.770 342	3.217 213	3.087 208	4.202 614	2.020 698	2.521 079	0.871 765	0.969 21
Basalt (Iceland) (BIR-1a)	1.051 139	0.816 578	2.428 145	2.574 803	3.108 512	1.741 674	1	1.957 997	2.946 852	2.787 844	1.529 582	1.262 963	0.891 713	0.751 164
Basalt (Deccan) (DVP)	0.056 809	0.055 736	0.068 268	0.085 05	0.136 68	0.521 325	0.163 525	0.208 804	0.247 288	0.261 53	0.269 612	0.288 558	0.293 478	0.309 725
Andesite (AGV-2)	0.099 649	0.079 266	0.067 829	0.065 562	0.069 479	2.349 037	0.069 752	0.079 076	0.089 326	0.097 753	0.110 299	0.122 851	0.130 962	0.136 307
Granite (Dh-26)	0.013 213	0.013 91	0.012 183	0.012 38	0.014 64	0.091 828	0.011 796	0.015 38	0.018 008	0.019 33	0.021 958	0.023 878	0.024 779	0.027 85

Table 4: Partition Coefficient (K_D) values of Major Elements

Partition Coefficient	Na	Mg	Al	Ca	Ti	Fe	Mn
Dunite	1.63378	0.415882	0.662182	3.084588	0.441077	0.411583	0.466667
Basalt (Iceland)	2.932379	0.301047	0.609756	5.808211	0.378523	0.303952	0.300658
Deccan (Deccan)	0.412174	0.518144	0.229122	0.456928	7.879485	0.226977	0.429603
Granite	9.850897	0.019473	2.652526	0.408958	1.860519	0.0243	0.028344
Andesite	3.381116	1.273758	5.175401	1.528868	4.418803	0.235208	0.393425